

Election Intelligence Analytics

A Data Analytics Capstone Project

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BUSINESS PROBLEM



Elections generate massive amounts of data.



Extracting meaningful insights is challenging.



Stakeholders need clear analysis of voting patterns, party performance, and regional trends.



This project aims to analyze election data to provide actionable insights.



PROJECT OBJECTIVES



Analyze overall election results



Compare performance of parties and candidates



Identify voting trends and patterns across states/regions



Evaluate winning margins and vote distributions



Generate actionable insights for better decision making



DATASET OVERVIEW



Source

Kaggle – Indian General Elections Dataset



Total Rows

~ X (example: 1,05,000+)



Total Columns

~ Y (example: 15+)



Key Features

State, Constituency, Candidate, Party, Votes, Margin, Result, etc.



Data Types

Integer, Float, Object



PROJECT WORKFLOW



DATA QUALITY ASSESSMENT & VALIDATION

The datasets were loaded and validated using Python (Pandas) and SQL to ensure data quality before analysis.

PYTHON – DATA VALIDATION (PANDAS)

Check Missing Values

```
isnull().sum()
constituency    0
ac_number       0
candidate       0
party           0
votes           0
turnout         0
reserved        0
region          0
dtype: int64
```

```
e3.isnull().sum()
constituency    0
ac_number       0
candidate       0
party           0
votes           0
turnout         4257
reserved        0
region          0
dtype: int64
```

Check Duplicate Records

```
duplicated().sum() # Duplicate count in 2021 data
duplicated().sum() # Duplicate count in 2026 data
```

Check Data Types

```
e3.dtypes
constituency    object
ac_number       int64
candidate       object
party           object
votes           int64
turnout         float64
reserved        object
region          object
dtype: object
```

SQL – DATA VALIDATION

-- 1. Check NULL values in each column (2021 Data)

```
SELECT
SUM(CASE WHEN constituency IS NULL THEN 1 ELSE 0 END) AS constituency_nulls,
SUM(CASE WHEN ac_number IS NULL THEN 1 ELSE 0 END) AS ac_number_nulls,
SUM(CASE WHEN candidate IS NULL THEN 1 ELSE 0 END) AS candidate_nulls,
SUM(CASE WHEN party IS NULL THEN 1 ELSE 0 END) AS party_nulls,
SUM(CASE WHEN votes IS NULL THEN 1 ELSE 0 END) AS votes_nulls,
SUM(CASE WHEN turnout IS NULL THEN 1 ELSE 0 END) AS turnout_nulls,
SUM(CASE WHEN reserved IS NULL THEN 1 ELSE 0 END) AS reserved_nulls,
SUM(CASE WHEN region IS NULL THEN 1 ELSE 0 END) AS region_nulls
FROM election2021;
```

constituency_nulls	ac_number_nulls	candidate_nulls	party_nulls	votes_nulls	turnout_nulls	reserved_nulls
0	0	0	0	0	0	0

-- 2. Check NULL values in each column (2026 Data)

```
SELECT
SUM(CASE WHEN constituency IS NULL THEN 1 ELSE 0 END) AS constituency_nulls,
SUM(CASE WHEN ac_number IS NULL THEN 1 ELSE 0 END) AS ac_number_nulls,
SUM(CASE WHEN candidate IS NULL THEN 1 ELSE 0 END) AS candidate_nulls,
SUM(CASE WHEN party IS NULL THEN 1 ELSE 0 END) AS party_nulls,
SUM(CASE WHEN votes IS NULL THEN 1 ELSE 0 END) AS votes_nulls,
SUM(CASE WHEN turnout IS NULL THEN 1 ELSE 0 END) AS turnout_nulls,
SUM(CASE WHEN reserved IS NULL THEN 1 ELSE 0 END) AS reserved_nulls,
SUM(CASE WHEN region IS NULL THEN 1 ELSE 0 END) AS region_nulls
FROM election2026;
```

constituency_nulls	ac_number_nulls	candidate_nulls	party_nulls	votes_nulls	turnout_nulls	reserved_nulls
0	0	0	0	0	4257	0

-- 3. Check Duplicate Records

```
SELECT COUNT(*) AS duplicate_count_2021
FROM (
    SELECT * FROM election2021
    GROUP BY ac_number, candidate, election_year
    HAVING COUNT(*) > 1
) t;
```

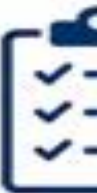
duplicate_count_2021
0

```
SELECT COUNT(*) AS duplicate_count_2026
FROM (
    SELECT * FROM election2026
    GROUP BY ac_number, candidate, election_year
    HAVING COUNT(*) > 1
) t;
```

duplicate_count_2026
0

OBSERVATION

- The datasets were successfully loaded and validated using Pandas and SQL.
- Duplicate records and data types were checked.
- All key fields such as candidate details, constituency, party, votes, and victory margin are consistent.
- For the 2026 data, all 4,257 turnout records were found to be NULL in SQL, so turnout-based analysis was excluded until the field is validated.



7 SQL BUSINESS ANALYSIS

SQL analysis generated the four key analytical views carried forward into the project story.

01

Largest Victory Margins

Decisive electoral outcomes

Slide 10

02

Closest Electoral Contests

Most competitive contests

Slide 11

03

Reserved Constituency Distribution

General, SC and ST representation

Slide 12

04

Top 10 Candidates by Votes

Candidate-level vote performance

Slide 13

SQL Analysis → Python EDA / Visualization → Business Insights

IMPORT SQL RESULTS INTO PYTHON



Connected Python
to MySQL



Executed SQL queries



Imported results
into Pandas DataFrame



Prepared data for
Exploratory Data
Analysis (EDA)



```
import pandas as pd
import mysql.connector

con = mysql.connector.connect(host='localhost', user='root',
password='*****', database='election_db')
df = pd.read_sql("SELECT * FROM election_analysis", con)
df.head()
```


EXPLORATORY DATA ANALYSIS (EDA)

Exploratory insights from Tamil Nadu Assembly Election dataset using **SQL** & **Python**



1. STATISTICAL SUMMARY

Dataset overview using info() & describe()



2. PARTY-WISE SEAT ANALYSIS

Ranking of political parties by total seats won



3. PARTY-WISE TOTAL VOTES

Comparison of total votes received by political parties



4. DISTRICT-WISE CANDIDATE ANALYSIS

Number of candidates contesting in each district



5. REGION-WISE CANDIDATE ANALYSIS

Distribution of candidates across regions



6. RESERVED CONSTITUENCY DISTRIBUTION

Distribution of constituencies by category (GEN / SC / ST)



7. AVERAGE VOTER TURNOUT ANALYSIS

- By Region
- By Election Year



8. LARGEST VICTORY MARGIN

Top constituencies with the highest victory margins



9. CLOSEST ELECTORAL CONTESTS

Constituencies with the smallest victory margins



Election Dataset



MySQL Database



SQL Queries

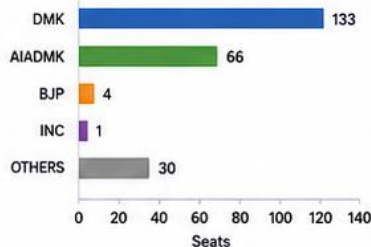


Python (Pandas)

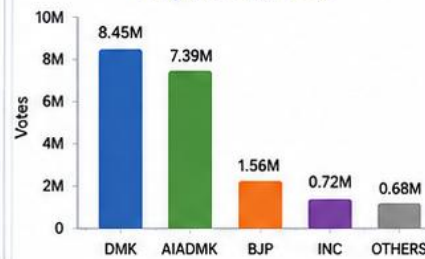


Insights & Visualizations

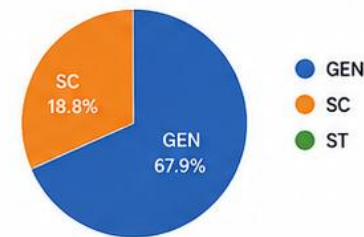
Party-wise Seats Won



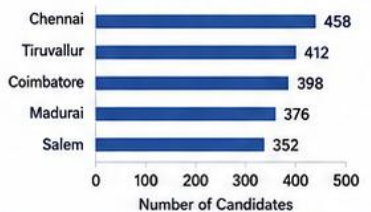
Party-wise Total Votes



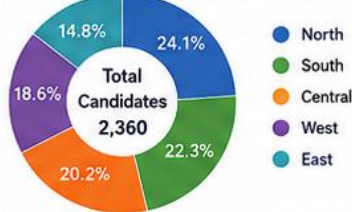
Reserved Constituency Distribution



Candidates by District (Top 5)



Candidates by Region



Average Voter Turnout (%)

By Region		By Election Year	
North	74.21%	2021	72.69%
South	72.85%	2016	74.52%
Central	71.68%	2011	70.48%
West	70.34%	2006	69.23%
East	69.12%	2026	0.00%

Top 5 - Largest Victory Margin



Constituency	Winner	Victory Margin
Chepauk	DMK	85,231
Mylapore	DMK	84,590
Egmore	DMK	82,441
Teynampet	DMK	78,932
Thousand Lights	DMK	76,521

Top 5 - Closest Electoral Contests



Constituency	Winner	Victory Margin
Perambur	DMK	842
Vikravandi	AIADMK	921
Sriperumbudur	DMK	1,102
Kilpauk	AIADMK	1,342
Arakkonam	BJP	1,523



OBJECTIVE: To explore, analyze and derive meaningful insights from Tamil Nadu Assembly Election data using SQL queries and Python data analysis.



OUTCOME: Actionable insights on party performance, voter behavior, regional trends and electoral competitiveness.



LARGEST VICTORY MARGINS

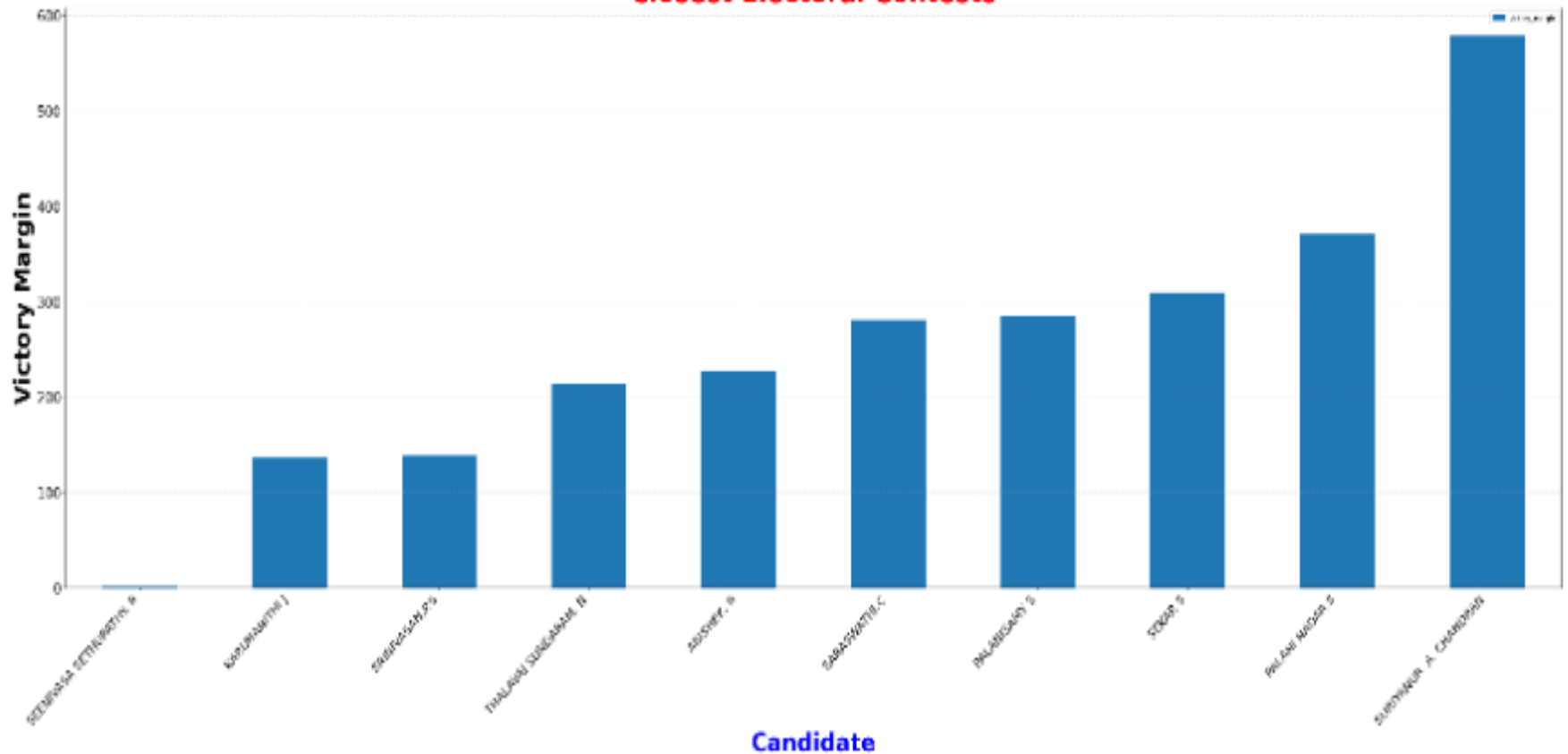


OBSERVATION

The chart highlights the 10 candidates with the largest victory margins. Periyasamy T has the highest victory margin among the candidates shown, while the remaining candidates also recorded substantial winning margins.

CLOSEST ELECTORAL CONTESTS

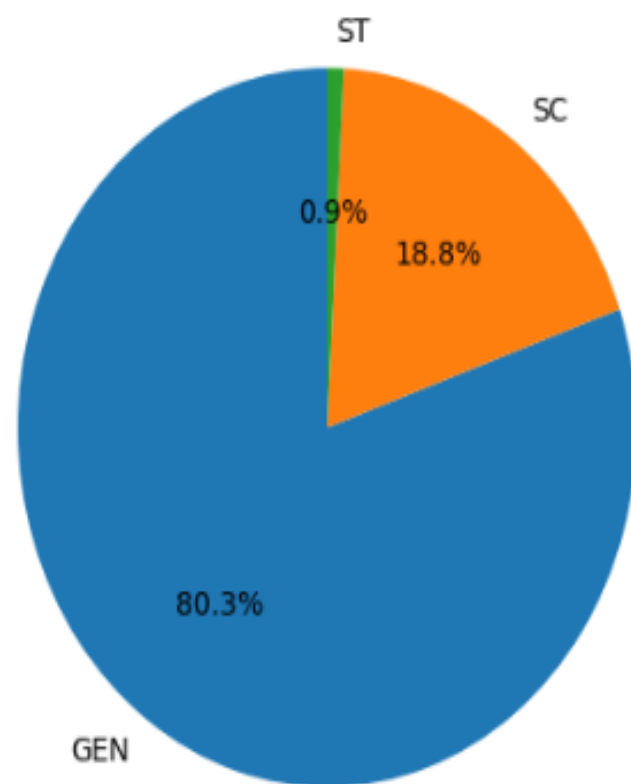
Closest Electoral Contests



OBSERVATION

The chart highlights the candidates with the smallest victory margins in the dataset. These results represent the closest electoral contests and provide a clear view of the most competitive outcomes.

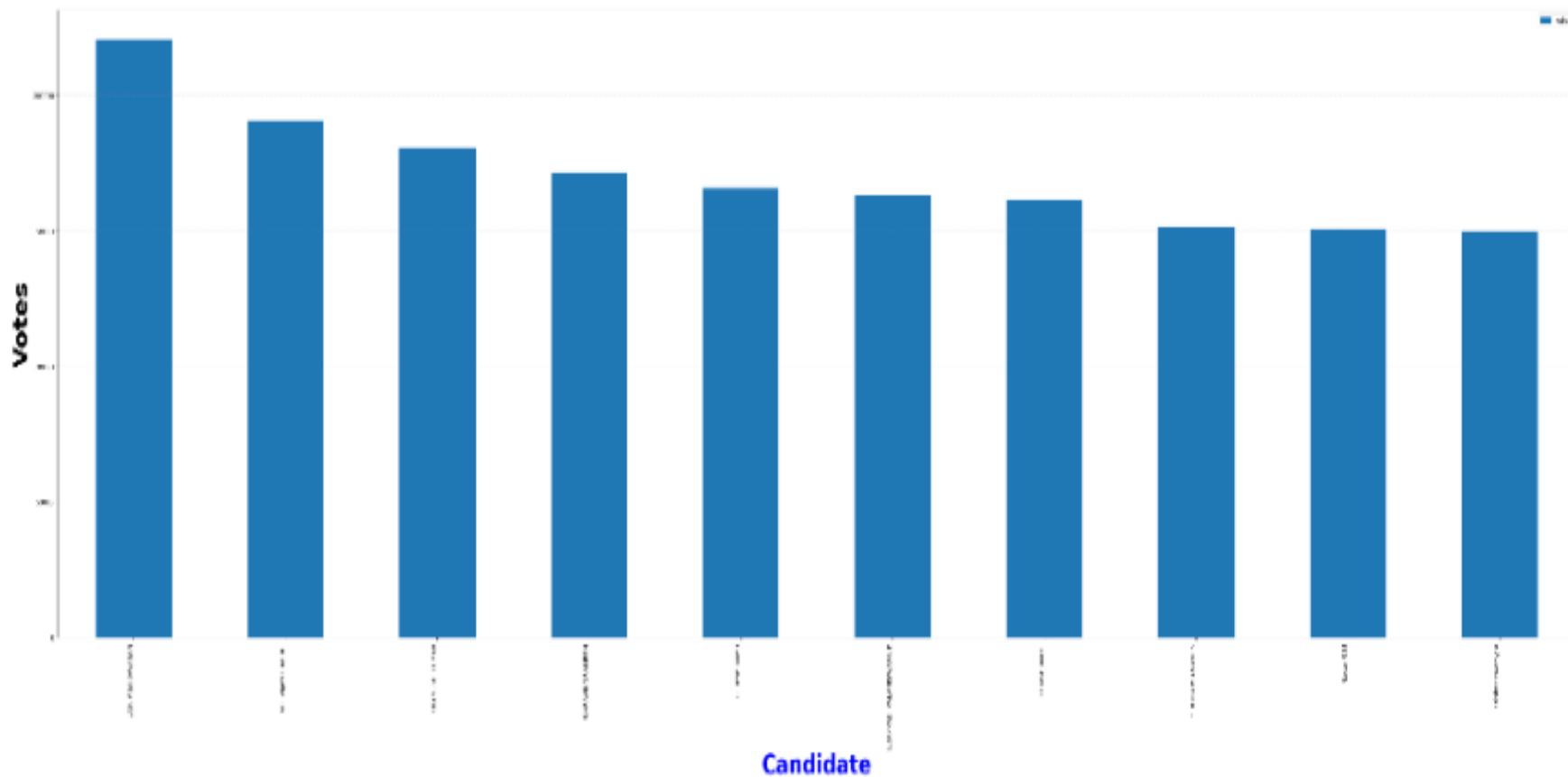
Reserved Constituency Distribution



OBSERVATION • General constituencies form the largest share (80.3%), followed by SC (18.8%) and ST (0.9%).

TOP 10 CANDIDATES BY VOTES

Top 10 Candidates by Votes



OBSERVATION • The chart highlights the candidates with the highest vote counts and provides a direct comparison of candidate-level performance.

SQL Analysis: Finding Closest Electoral Contests



election_year	ac_number	candidate	party	victory_margin
2026	185	SEENIVASA SETHUPATHY, R	TVK	1
2021	24	KARUNANITHI J	DMK	137
2026	54	SRINIVASAN,P.S	DMK	138
2026	229	THALAVAI SUNDARAM, N	AIADMK	214
2026	66	ABISHEK, R	TVK	227
2021	100	SARASWATHI,C	BJP	281
2026	76	PALANISAMY S	AIADMK	285
2026	95	SEKAR S	AIADMK	308
2021	222	PALANI NADAR,S	INC	370
2026	137	SURIYANUR, A. CHANDRAN	DMK	579

OBJECTIVE

Which constituencies had the closest competition?

KEY OBSERVATIONS

- Ranked candidates based on votes.
- Compared winner and runner-up votes.
- Calculated minimum victory margins.
- Identified highly competitive constituencies for deeper analysis.

SQL Analysis: Identifying Electoral Performance

-- 5.1 Largest Victory Margin

```
WITH ranked AS (  
    SELECT *, ROW_NUMBER() OVER (PARTITION BY election_year, ac_number  
        ORDER BY votes DESC) AS rn  
    FROM election_results),  
margins AS (  
    SELECT w.election_year, w.ac_number, w.candidate, w.party, w.votes -  
        (SELECT votes FROM ranked r  
        WHERE r.election_year = w.election_year  
        AND r.ac_number = w.ac_number  
        AND r.rn = 2) AS victory_margin  
    FROM ranked w WHERE rn = 1)  
SELECT  
    election_year, ac_number, candidate, party, victory_margin  
FROM margins  
ORDER BY victory_margin DESC LIMIT 10;
```

election_year	ac_number	candidate	party	victory_margin
2021	129	PERIYASAMY I	DMK	135571
2026	86	EDAPPADI PALANISWAMI. K	AIADMK	98110
2026	27	ECR P SARAVANAN	TVK	96780
2026	9	M.L.VIJAYPRABHU	TVK	94985
2021	63	E V VELU	DMK	94673
2021	5	Krishnaswamy A	DMK	94110
2021	86	EDAPPADI PALANISWAMI. K	AIADMK	93802
2021	140	Nehru, K.N.	DMK	85109
2026	6	R.RAMESH KUMAR	TVK	76311
2026	88	LAKSHMANAN.S	TVK	74867

Which candidates achieved the largest victory margins?

KEY OBSERVATIONS

- ROW_NUMBER() ranked candidates within each constituency; winner and runner-up votes were compared to calculate the victory margin.

KEY FINDINGS & INSIGHTS

Evidence-based findings from the SQL analysis and Python EDA.

01

Party Performance

Party-wise analysis revealed differences in total votes and seats won.

02

Candidate Performance

Candidate-level analysis highlighted the highest vote-getters and decisive victo

03

Electoral Competition

Victory-margin analysis identified both highly competitive and decisive

04

Constituency Representation

Reserved constituency analysis showed the distribution of constituency categ

DATA VALIDATION NOTE • The 2026 turnout field requires source validation before turnout-based interpretation.

RECOMMENDATIONS



Increase voter awareness programs in low turnout regions.



Focus on key issues affecting voters in competitive areas.



Use data driven strategies for campaign planning.



Monitor historical trends for better future predictions.



TOOLS & TECHNOLOGIES



Python



Pandas



NumPy



Matplotlib



Jupyter
Notebook

CHALLENGES & LEARNINGS

CHALLENGES

- ✓ Understanding large and complex dataset
- ✓ Writing optimized SQL queries
- ✓ Managing relationships between tables
- ✓ Presenting findings in a meaningful way



LEARNINGS

- ✓ Data validation using Pandas
- ✓ SQL query writing and optimization
- ✓ Exploratory Data Analysis
- ✓ Data visualization
- ✓ Data storytelling & deriving insights



CONCLUSION

- ✓ The project demonstrates end-to-end analysis of election data.
- ✓ SQL helped in extracting meaningful information.
- ✓ Python was used for in-depth EDA and visualizations.
- ✓ Insights derived can support data driven decision making.

Thank You!

Questions & Discussion

